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Analysis and Detection of LLM-Generated Text in Swedish

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Department of Linguistics and Philology
Master Programme in Language Technology
Master's Thesis in Language Technology, 30 ECTS credits

May 6, 2026

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Abstract

This thesis investigates investigate systematic differences between text generated by large language models (LLMs) and human-written text in the Swedish language. In addition, the thesis analyses significant differences in the ability to detect LLM-generated Swedish text between humans and AI-detection tools. The study will include both a formal and informal text component, that is, linguistic analyses and AI-detection experiments will be performed on two sources of data from formal and informal domains.

Contents

Preface	4
1 Introduction	5
1.1 Research questions	5
1.2 Experimental-set up of the thesis	6
2 Background	7
2.1 Transformer-based Large Language Models	7
2.2 LLM development methodology	7
2.3 LLM textual analysis studies	9
2.4 LLM-detection studies	10
2.4.1 LLM-detection by humans	10
2.4.2 LLM-detection by LLMs	10
2.4.3 LLM-detection by LLMs	11
2.5 Background summary	11
3 Methodology	12
3.1 Human-written data collection	12
3.1.1 Human-written abstracts	12
3.1.2 Human-written internet comments	12
3.2 LLM data generation	13
3.3 Abstract data	13
3.3.1 LLM-generated informal data	13
3.4 Experiment 1: textual analysis	14
3.4.1 Linguistic structural analysis	14
3.4.2 Content analysis	14
3.5 Experiment 2: LLM-generated text detection	14
3.5.1 LLM-detection experimental set-up	14
3.5.2 Human detection	14
3.5.3 AI detection	15
4 Results	16
4.1 Textual analysis	16
4.1.1 Abstracts	16
4.1.2 Comments	17
4.2 AI detection	17
5 Discussion	18
5.1 Textual analysis results	18
5.2 AI detection results	18
5.3 Limitations	18
6 Conclusion	20

Preface

I would like to thank the following people...

1 Introduction

The invention of transformer-based Large Language Models (LLMs) has created a new wave of textual data that is entirely generated by machines and very closely emulates, to the naked eye, text written by humans. As these tools continue to generate unbelievable amounts of textual data at a break-neck speed (« insert some figure taken recently on number of TB per year »), changing the landscape of data proliferated through the internet and in other published textual sources, such studies could also shed light on what kind of text is being read by the literate population, which styles are dominating, and what kind of language structures and content is being prioritised through the usage of these generative AI. In addition, studies such as these can inform readers of potential heuristics to use when reading a text, influencing their intuition to scrutinise a text and potentially infer their origin from a machine. Analysing the characteristics of LLM-generated data enhances the ability to shed light on the mechanisms of LLMs, potentially contributing to transparency and LLM-probing research within the NLP space.

In addition, there has been emerging research into detection methods of LLM-generated text, both by AI detection tools and by humans. These studies help with development of AI detection tools and to further educate the general public in building their intuition of LLM-generated text. There is recent evidence that LLM-generated text has changed the make-up of text on the internet (INSERT SOURCES), with an estimated (SOURCE) % of the internet estimated to be made up of LLM-generated text (although this figure is only estimated as it is difficult to gauge). AI-detection tools are useful in several contexts, including plagiarism detection and the propagation of LLM-generated text containing political propaganda and hate speech to influence users on the internet. These topics are pressing matters due to the number of users on the internet and the risk of political order being compromised and bias against groups being reinforced with the help of LLM tools, which can create massive amounts of data very quickly.

The study of the linguistic properties of LLM-generated text has emerged from this recent phenomenon as a growing topic of natural language processing (NLP). Recent research has focused primarily on English-language textual analysis as the dominant type of data analysed. However, there has been thus far limited but emerging research into the characteristics of LLM-generated text in languages other than English, such as Swedish. Hence, there is a lot of opportunity for exploration of this topic in many other languages.

1.1 Research questions

The intended purpose of this study is to investigate systematic differences (if any) between text generated by large language models (LLMs) and text written by humans in the Swedish language. In addition, the study will investigate whether there are any significant differences in the ability to detect LLM-generated Swedish text between humans and LLM tools. The study will include both a formal and informal text component, that is, linguistic analyses and AI-detection experiments will be performed on two sources of data from formal and informal domains.

1. Are there systematic differences between LLM-generated and human-written formal text in Swedish?
2. Are there systematic differences between LLM-generated and human-written informal text in Swedish?
3. Are there differences between humans and AI-based detection tools in their ability to detect LLM-generated Swedish text?

1.2 Experimental-set up of the thesis

The thesis is comprised of two experiments: 1) textual analysis, and 2) LLM detection. The first experiment focuses on how human-written and LLM-generated text differ in surface-level and content-level characteristics. There are two domains of text used: 1) a formal domain comprised of Bachelor's level thesis abstracts, and 2) an informal domain of internet thread comments. The LLM used to generate the texts is OpenAI's GPT-5.2. The second experiment focuses on the differences in ability and thinking process between humans and LLMs in detecting LLM-generated text. The LLM of focus is DeepSeek (? plus model version), and the accuracy and reasoning process will be evaluated for both Swedish-native speakers and this LLM in detecting both the formal and informal data.

2 Background

This chapter provides an overview of previous studies related to the topics of LLM development, textual analysis, and detection. First, it will outline the development of large language models and their text generation abilities for English and Swedish. Second, it will describe previous research comparing AI-generated text with human-written text. The chapter will then outline tools and human participant studies related to AI-generated text detection.

2.1 Transformer-based Large Language Models

[Purpose: introducing the type of language model referred to in thesis, some architectural characteristics, and cross-task abilities, NEED ADD CITES]

In this thesis, the term "Large Language Models" refers to those that are transformer-based, that is, based on a neural network architecture with multi-head attention mechanisms. This type of architecture was first proposed in [vaswani2017attention](#), where the authors developed an architecture that did not rely on earlier recurrent neural architectures and demonstrated superior performance and lower training time for a variety of tasks, including text generation, including optical recognition, speech processing, and text generation (CITE Islam 2023, Radford). The strength of transformers in their adaptation to a wide variety of tasks lies in their ability to handle long dependencies between different elements of input sequences, meaning that it can calculate how relevant different parts of input are to one another, no matter how far apart one element is from another within a sequence. This ability has shown to improve the fluency and coherence of text, as well as emergent abilities that are not found in earlier language models, such as instruction-following (CITE) and reasoning (CITE). Examples of transformer-based large language models include GPT-5 (GET PAPER 5; <https://arxiv.org/pdf/2303.08774v3>), Claude Sonnet (GET PAPER), and LLaMa (GET PAPER). LLMs are autoregressive, meaning that it predicts the next word of a sentence based on all words it has generated up until that point as context.

2.2 LLM development methodology

[Purpose: Detailing the parameters, data sources of LLMs; training process]

Compared to earlier statistical and neural network models, LLMs that have been released after GPT-3 are trained on very large amounts of data. Transformer-based LLMs for text generation normally contain tens to thousands of billions of parameters, pre-trained on diverse sources of textual data such as Web crawl data (CITE), online encyclopedias, news outlets, academic articles, books, and code repositories. The scale of data and parameters varies by each LLM and release version. During model development, the data collected is then often filtered to enhance data quality, relevance and generalisation of the data to different user inputs (CITE1, CITE2, (Brown2020LanguageMA)).

LLMs and the Swedish language

Several studies have developed datasets, models and evaluation methods specifically for Scandinavian languages. The most notable dataset is the Nordic Pile (Öhman et al., 2023), a 1.2 terabyte dataset trained primarily on Swedish, Norwegian, Danish, Icelandic, Faroese, and smaller subset of high-quality English data. The Nordic Pile was used to train GPT-SW₃ (Ekgren et al., 2024), the first Scandinavian GPT model, the largest largest release model of which has 40 billion parameters at the time of writing. Further studies that incorporate GPT-SW₃ in their pipelines include Olsén (2024), which focuses on instruction-tuning for Swedish language tasks, and (INSERT ANOTHER GPT-SW₃ STUDY). AI Sweden has also released a model that further pretrains Llama (cite) with a subset of the Nordic Pile (footnote <https://huggingface.co/AI-Sweden-Models/Llama-3-8B>).

Existing evaluation benchmarks that are specifically tailored for LLM-generated Swedish include ScandEval (Nielsen, 2023), which offers several metrics including Swedish named entity recognition (NER), sentiment classification, and linguistic acceptability; a Sweden-related factual knowledge benchmark dataset by Kunz (2025); SweSat (Kurfali et al. 2025), a benchmark from the Swedish university entrance exam (Högskoleprovet); and the Swedish Medical LLM Benchmark (Moëll et al. 2025), a framework to evaluate LLM capabilities for Swedish medical texts. These benchmarks can be evaluated on large-scale multilingual LLMs to test their performance specifically for the Swedish language and cultural context.

Multilingual LLMs

[Purpose: this thesis uses a multilingual LLM, need to know what cross-lingual capabilities transfer to Swedish, and what limitations it has. Cultural limitations? To guide the discussion of how the English-centric LLM could have limitations in the syntax and cultural information of the Swedish performance and AI detection reasoning.] Latest-release, large-scale LLMs such as OpenAI’s GPT, Anthropic’s Claude, and Google’s Gemini are trained on data from multiple languages and therefore have multilingual capabilities (CITE OA, CITE AN, CITE GE). It is difficult to ascertain the exact amount of linguistic diversity within the training data of large-scale multilingual LLMs, as there is no publicly-accessible, detailed breakdown of the languages represented in the training data of the latest versions of large-scale multilingual LLMs, such as OpenAI’s GPT, Anthropic’s Claude, and Google’s Gemini (check more open source ones?).

However, an approximation of large-scale multilingual LLM’s linguistic diversity can be obtained through other openly-accessible sources. The original paper that proposed the GPT-3 autoregressive language model (CITE; Brown et al. 2020) released a dataset detailing its multilingual distribution; 92.6% of the total training data’s word count was in English; the next languages were The Swedish word count proportion was 0.11%. In addition, the Common Crawl monthly archives, an open repository of web data cited as a major source in several large scale LLMs (CITE₁, CITE₂, CITE₃), calculates the percentage of Swedish data within its corpus as between 0.68 - 0.72 % across three versions released in 2026 (<https://commoncrawl.github.io/cc-crawl-statistics/plots/languages>). Overall, these sources indicate a proportion of Swedish training data within the large-scale, English-centric LLMs to be well below 1%.

Despite the asymmetrical language proportions in their data, LLMs display emergent multilingual properties in pattern recognition of the linguistic structures of supported

languages, displaying a high level of fluency across a range of natural language understanding and generation benchmarks (cite: Kew et al., 2024; Lai et al, 2023; Holmstrom et al., 2023; need more recent cites). On the other hand, since LLMs embed cultural norms and biases inherent in their training, they require benchmarking of not only linguistic fluency, but also cultural alignment - that is, the cultural references, expectations and biases of non-Anglocentric cultures. While there is also considerable evidence that LLMs can in fact successfully reflect the cultural contexts of non-Anglocentric cultures, some show that LLMs' performance are still stronger in English than other languages for reasoning tasks (Liu et al, 2024), figurative proverbs (Liu et al., 2024), and (insert other cites with tasks).

"Multilingual considerations demand increased attention. Specifically, existing benchmarks are mainly developed for English datasets, with only a few like MULTI-TuDE (Macko et al. 2023) and M4 (Wang et al. 2023b) covering multiple languages, including Arabic, Catalan, Chinese, Czech, Dutch, English, German, Portuguese, Russian, Spanish, Indonesian, Bulgarian, and Ukrainian. However, there is still a significant lack of datasets for other language resources, especially low-resource languages." - Wu et al, 2024

- need to add all studies that include the Nordic pile as a source.

LLMs and cultural transfer

Given the dominance of English-centric multilingual LLMs, emerging result into the cultural transfer cultural transfer of English-centric multilingual LLMs to other languages smaller languages like Swedish, spoken primarily in a geographically smaller area (in the case of Swedish: Sweden, Finland). Holmström et al. (2023) examined the cultural adaptation of English-centric pre-trained GPT-3 LLMs to the Swedish context. Their findings indicated that despite LLMs' exposure to limited Swedish during training, they demonstrate a high capability to culturally adapt to the Swedish context "off-the-shelf" without unwanted over-generalisation of Swedish cultural aspects.

2.3 LLM textual analysis studies

[Purpose: give a breakdown of previous studies that have textually analysed the output of LLMs, or put together signs of LLM-generated writing. First start with English (and other languages), then focus on Swedish studies]

These studies are focused on increasing transparency of signs of AI writing for the general public. (need more introduction).

- "Additionally, LLM-generated text often conveys less emotional intensity and exhibits clearer presentation than human writing. This phenomenon is possibly influenced by inherent positive bias in LLMs (Giorgi, Ungar, and Schwartz 2021; Markowitz, Hancock, and Bailenson 2023; Mitrović, Andreoletti, and Ayoub 2023)"

Munoz-Ortiz et al. (2024) compared the linguistic components of human and LLM-generated news articles and compared their patterns among six generative LLMs. For their prompting strategy, they asked the models to generate a news paragraph based on the news article headline its first words. For their linguistic analysis, they compared the human-written and LLM-generated text groups through differences in their linguistic structures, including vocabulary distribution, part-of-speech tag distributions,

dependency arc lengths, constituent lengths, cosine similarity, and emotional category tagging. The results of this analysis showed significant differences between the human-written and LLM-generated text on all textual analysis metrics. A sentiment analysis of the models found less of an inclination to display aggressive negative emotions (such as fear or anger) than humans. This finding could be influenced by inherent positive bias in LLMs (Wu et al., 2024; Giorgi et al., 2021; Markowitz et al., 2023), which be a result of safety-policy filtration of negatively-biased data within LLM training (CHECK - Liao et al., 2023?).

Liao et al. (2023) compared human-written and ChatGPT-generated medical texts by their vocabulary distribution, part-of-speech tag distribution, dependency arcs, sentiment analysis, and perplexity. Their results found that human-written medical texts contained more concrete information, had higher vocabulary diversity, and contained information that human experts would consider more useful. The ChatGPT-generated medical texts appeared more fluent and logical, but expressed more generalised terminology rather than specific, effective information for the problem's context, which could be attributed to human-written medical text being more tailored to situational modelling of the particular medical case (Mahowald et al., 2024).

- need add more synthesis

2.4 LLM-detection studies

2.4.1 LLM-detection by humans

[need add more info; Wikipedia signs of AI writing referenced studies]

2.4.2 LLM-detection by LLMs

Landberg (2024) investigated the effectiveness of AI chatbots to detect texts created by AI in Swedish and English.

LLM reasoning

Previous work has also shown evidence of a distinction between the reasoning patterns of LLM tools when compared to those of humans. (CITE) evaluated the medical reasoning capabilities of DeepSeek R1, computing its outputs to the reasoning patterns of medical domain human experts. The study performed qualitative and quantitative analyses of AI- and human-medical reasoning for 100 clinical cases from the MedQA dataset. The study found a 93% diagnostic accuracy for DeepSeek R1's performance. Many of the LLM's errors were attributed to over-focus on the details of a medical problem, rather than a more holistic and intuitive (?) understanding by humans, as well as an insufficient understanding of medical protocols when compared to human experts. The study also found a strong correlation between the length of output and the error rate of LLMs; that is, results from LLMs that were longer were more likely to contain incorrect diagnoses or medical details.

Möell and Boye (2025) studied the performance of LLMs on language complexity measurement tasks (LIX readability metric and Average Dependency Distance). The results of the study found that all models demonstrated at least some capacity for each of these tasks, but that LLMs remain prone to errors in explicit calculations and syntactic analysis. [need add more info; Wikipedia signs of AI writing referenced studies]

2.4.3 LLM-detection by LLMs

- need more studies here!!

2.5 Background summary

[insert summary]

3 Methodology

– wait wtf?? why are there 2-week old questions that are having LLM generated comments off it??

The methodology is comprised of two main parts: 1) a comparison of textual elements between human-written Swedish and LLM-generated Swedish, and 2) a comparison between the detection rates of humans and LLM-detection of LLM-generated Swedish. Both parts of the methodology comprise of a human component and an LLM-generated component.

- why choose GPT-5.2 multilingual instead of SW3? GPT-5.2 was one of the top k performers for the Swedish language (CITE), with a massive estimated number of parameters (cite number, source). SW3, on the other hand, only had x number of parameters. The study did not want to have number of parameters as a confounding variable for the results of the study, and also didn't want to compare between LLM architectures without having a large-scale amount of Swedish human-written data

3.1 Human-written data collection

The data collection of this thesis formed the data source of both the textual analysis and LLM detection experiments. The dataset comprises of 1) a set of texts in a formal register, collected through abstracts; and 2) a set of texts in an informal register, collected through forum comments. These two registers are selected to measure differences in textual elements and detection rates of LLM-generated text within two different contexts.

3.1.1 Human-written abstracts

The formal human-written dataset comprises of (n=142(?)) Bachelor's level thesis titles, abstracts and keywords sourced from the Digitala Vetenskapliga Arkivet (DiVA) portal. All Swedish abstracts and keywords were written originally in Swedish. All abstracts were sourced between 2007-2017, to negate the chance of of any transformer-based generative models influencing any part of the dataset's titles or abstracts. The topics of the thesis abstracts were sampled randomly across (insert number) topics. On average, the thesis abstracts were (n=words) in length.

3.1.2 Human-written internet comments

– THE DATA COLLECTION FOR THIS NEEDS TO BE RE-DONE

The informal human-written dataset, to measure the Swedish language in a more casual register, was comprised of 203 comment replies to (insert n=questions) Swedish-language Reddit questions, retrieved across subreddit forums r/Sweden, r/svenskpolitik, r/Stockholm and r/Goteborg. The average length of comments was (n=comment length). To mitigate the possibility of transformer-based generative AI tools being involved in the creation of the subreddit's questions and replies, only questions and comments with a timestamp before January 2017 (2017-01-01) were scraped via the PullPush API (cite or footnote). To ensure that the post's titles were in question form

and in Swedish, the scraper script checked for the presence of a question mark ("?",) and used the <langdetect> Python library to identify the post as being written in Swedish ("sv"). On average, the Reddit comments

3.2 LLM data generation

All LLM-generated data for this study was created by OpenAI's GPT-5.2 multilingual model (cite). GPT-5.2 was selected for this study, since at the time of writing, this model was one of the top k performers for the Swedish language (CITE), with a estimated number of parameters (n = cite number, source). GPT models are also widely used by the community, with the number of users estimated as (n=n), which would more closely simulate a real-life scenario of using LLMs to generate text. AI Sweden's SW3 is trained on Scandinavian data; however, the model at the time of writing was trained on x number of parameters, making it incomparable to the widely-used large-scale models (GPT, Claude, Gemini). The study did not want to have the number of parameters as a confounding variable for the results of the study. The study also chose one model as the source of text generation, since a systematic comparison of the results between different LLM architectures without a larger-scale amount of Swedish human-written and generated data would have reduced the statistical power of the results.

3.3 Abstract data

The formal human-written dataset comprises of an LLM-generated version of each human-written sample of the DiVA Bachelor's-level thesis abstracts. The model used was OpenAI's GPT 5.2 model (<gpt-5.2>). The prompt used as input to generate the comments was:

[INSERT BREAKDOWN OF GPT 5.2 DATA]

<https://arxiv.org/pdf/2601.03267> - trained on diverse datasets,

<https://apxml.com/models/gpt-52-high>

-

"Titta på titeln och nyckelorden och skapa en sammanfattning i kandidatuppsatsstil med dina egna ord: <title, keywords>."

English translation: Look at the title and keywords and create a summary of the thesis abstract in your own words: <title, keywords>

- The LLM generation prompt also did not use additional evasion tactics such as "write it in the style as close as possible to a native Swedish speaker" or "use a more diverse amount of vocabulary". While these would be very interesting and useful experiments in further work on this topic, this study focused on a minimal prompt to control for evasion tactics, and to focus on the "bare-bones" / zero-shot generation of the Swedish text.

3.3.1 LLM-generated informal data

- prompt needs to remove word limit

"Svara på följande fråga med en svensk kommentar, i stil med en kommentar på ett svenskt forum. " "Skriv bara kommentaren, inget annat. Fråga: <question>"

3.4 Experiment 1: textual analysis

The aim of the textual analysis experiments is to investigate meaningful differences between LLM-generated and human-written Swedish. This method is based on the hypothesis that there are indeed meaningful differences in the writing style between the two textual sources in Swedish. The method aims to cover both surface-level analyses of the text, that is, investigating structural differences between the word forms and surface-level composition between the two modes; as well as content-level analysis, that is, sentiment analysis, entities, and themes of the texts.

3.4.1 Linguistic structural analysis

For the surface-level analysis, the spaCy package was used (sv_core_news_sm) to collect distributions of: vocabulary distribution, part-of-speech distribution, dependency analysis (sprakbanken?), dependency arch length, average word length, average sentence length.

3.4.2 Content analysis

- named entity recognition (spacy sv) - sentiment analysis (sentiment model 'KBLab/robust-swedish-sentiment-multiclass'.)

- why not thematic clustering to uncover both recurring ideas and themes through the abstracts and comments

For the content analysis, the study used the sentiment analysis, named entity recognition

3.5 Experiment 2: LLM-generated text detection

3.5.1 LLM-detection experimental set-up

Another LLM (DeepSeek or whatever) to do AI detection - quantitative measure (accuracy), qualitative measure (reasoning)

3.5.2 Human detection

For the human detection experiment, a web game was created by the author with 15 Reddit-style forum threads with 4 comment replies per question, and 15 thesis abstract questions. The URL of this web game was shared in Swedish-speaking forums, with the only prerequisite screening for users of the game was native-level knowledge of Swedish.

Before the game started, the user was prompted to fill in information about 1) their age, 2) their occupation, and 3) the estimated number of hours of their LLM tool usage per week. Each thread and abstract was shown one at a time, with instructions given to the user to determine whether each comments or abstract was LLM-generated or human-written. The proportion of AI comments or abstracts to human-written was sampled at random. After each question was completed, feedback was given immediately to the user as to whether their response was correct or not before moving on to the next round. This immediate feedback characteristic was decided in order to increase the chance of engagement by the user until the very end. Each user was given the option to exit the game at any time. At the end of the game session (or upon exit), the user was shown their accuracy results and prompted to fill in an optional short reflection to explain their reasoning process for LLM-detection throughout the game.

There were (n=?) game sessions submitted through the web game, each of which was stored in a Google Sheet form with information about their usage time-stamp, question identification number, which questions were determined as human, and accuracy scores. The results were measured in terms of accuracy (F-score too?), and qualitative analysis of any reasoning descriptions sent through the game.

3.5.3 AI detection

- the LLM detection prompt did not mention that the text was generated by GPT, even though this is mentioned as a potentially useful prompt in Wu et al. 2024. This was to closely simulate a real-life scenario where an LLM check is being used on online data, where the original model used to generate the text would not be known.

4 Results

[insert intro to results here]

4.1 Textual analysis

4.1.1 Abstracts

Part-of-speech categories

Punctuation: LLM has 2 percent more than human; Human-written abstracts have 7.5%, LLM-written has 9.5%; (llm written has about 2 per cent more) Case: human has 2 percent more than LLM; Formal human has about 11.5 percent, llm has about 9.5 (human 2 percent more) Adverbial modifiers: human has 2 per cent more than LLM Nominal subj: human has 3 per cent more than LLM Conjunction: llm has 5 percent more than human Adverbial modifier: llm has 0.5 percent more than human Coordinating conjunction (cc) - LLM has 4 percent more than human Oblique: human has 1 percent more than LLM Mark: about the same Obj: about the same Determiner: human has 2.5 percent more than LLM

Dependency categories

Named entity distribution

annotation guidelines from: <https://sprakbanken-clarin.lingfil.uu.se/rapporter/SCR-01-2020.pdf>

LOC = Locations MSR = Measure / Numerical expression ORG = Organization PRS = Persons TME = Time WRK = Works of Art and Other Artefacts

Across the board, humans had more entity types than LLM abstracts, indicating that the human-written abstracts are more inclined to concretely name people, organisations and geographic locations than the LLM abstracts. In particular, human-written had almost three times as many Location (LOC) entities as the LLM abstracts, indicating that human-written abstracts were more likely to name geographic cities, countries, and natural landmarks (e.g. "Stockholm", "Sverige", "Kungsgatan 24"). Human abstracts had more than twice the amount of Organization (ORG) entities as the LLM abstracts, indicating that they were more likely to name companies (e.g. Volvo), institutions (e.g. "Riksdagen"), and sports teams (e.g. "Malmö FF"). In addition, they named more than twice the amount of Persons (PRS) entities such as historical and cultural figures (e.g. "Astrid Lindgren", "Carl Linnaeus"). Human abstracts were 3 times more likely than LLM abstracts to denote Temporal entities (TME), that is, entities that cover points and continuous intervals on a timeline (e.g. "2 april 1991", "1800-talet", "Påskdagen", "idag").

Overall, it appears that human abstracts make more explicit references to entities within the world than LLM abstracts, indicating a more concrete and specificity in their treatment of academic topics than the LLM outputs. This finding is in accordance with

English-centric studies such as (CITE 1) and (CITE 2), which measures that English LLMs are more vague than human-written [EXPAND].

Sentiment analysis

Sentiment analysis: for Swedish academic abstracts, the vast majority were labelled as neutral, but formal LLM generated texts have slightly more positive sentiment. None of the abstracts for either group were labelled as negative

4.1.2 Comments

Dependency - Punctuation: formal LLM had about 4% more than human comments - Adverbial modifier: about the same - Nsubj: human slightly more than LLM - Conj: LLM about 5% more than human (cc: 3 percent more) - Case: human slightly more - Mark: human slightly more - Det: human about 2 percent more

Again like the abstracts, LLM-generated comments in Swedish have considerably more conjunctions than human comments (5 percent)

NER entity - Measured by mean entities per 100 tokens - LOC: human more (0.8 vs 0.7) - ORG: llm more (0.18 vs 0.21)

POS distribution

- NOUN: LLM more than human (3 percent more) - Punct: LLM more than human (3 percent more) - Pronoun: human more than llm (2.5 percent more)

Sentiment analysis - LLM comments were more neutral, very slightly more positive
- Humans were considerably more negative than LLM by about 10%

4.2 AI detection

- trained neural network-based detectors - zero shot detectors - watermarking - retrieval-based detectors

5 Discussion

5.1 Textual analysis results

Textual analysis:

5.2 AI detection results

- potential points of discussion:

5.3 Limitations

- Category was completely human written and completely LLM written, without mixed text examples, which is a plausible scenario (Wu et al., 2024, more)

- Human-written data was taken between 2007 - 2017, which did guarantee that no examples had any transformer-based LLM as help; however, human-written text written then could have outdated information or cultural references that mark the text as older and may be not so comparable for studies that use recent human-written data as examples, and does not match the AI text so much.

- Another issue that has not been fully discussed is the human-edited text and LLM-assisted text. For example, after applying LLM to generate a text, humans often edit and modify certain words or passages (Wu et al., 2024). Conversely, humans may write an original text themselves before employing LLMs to modify the text's content and structure. Current mainstream LLM detectors show evidence of degrading in performance when categorising paraphrased text (Wolff 2020). This study only used generation techniques that completely generate a passage themselves using the title and key words (not re-writing the human written examples), but this technique would be less comparable for studies that look at partially LLM-assisted text. However, this type of study would be interesting to conduct in future research.

- this study did not test the results of prompts that use evasion techniques, that is, techniques that change the output of an LLM in order to explicitly avoid detection, such as designing the prompt to give a more "human-like" response or other styles that suit the targeted domain. Instead, this thesis wanted to give a foundational result that future results could build on in the future. However, this would be very interesting to explore in future work.

- participants in the AI study gave feedback that it was easy to spot patterns in AI text as the game went on, since they were receiving feedback after each round which AI and which human. this could make the study contrived in a way, since humans are explicitly looking for feedback and able to spot patterns as time goes on. at the same time, the immediate feedback signal from the game aimed to incentivise participants to complete the whole study through motivation. so, there is a trade-off - have more non-paid participants engage in the game at the expense of results potentially being

skewed due to immediate feedback facilitating pattern recognition. Also, there was a relatively high drop-out rate in the quiz - many participants completed the whole comments section, but fewer participants finished the abstract section. So, there is more data points for the comments and not so much for the abstracts.

6 Conclusion

text here

bibliography file is empty